A Calibration of the standard parameters

My full calibration contains many parameters, many of which are more conventional and used in many papers. Here, I show how I set those thirty-three parameters. I calibrate six of them internally and set the other twenty-seven externally from estimates in the literature or from statute. Table A1 and the entries below proceed in the same seven blocks. In each entry, I give the moment or source that pins down the parameter and the value of that moment in the data. For the internally calibrated parameters, the Model column also presents the model's performance.

None of these parameters are inert. They too matter for the experiments, but they are more standard and matter less for the results than the parameters that I covered in the main text. For instance, the discount factor affects how much weight households give to dying in a later period. The dispersion of productivity determines how many households are poor enough to be willing to risk having a disorder, and risk aversion determines how much households value the insurance of housing assistance.

Table A1: Standard parameters

Parameter	Description	Set	Moment and source	Data	Model	Value
Block 7: Preferences						
σ	Risk aversion	Ext.	Attanasio and Weber (2010)	2.00	n.a.	2.00
γ	Weight on housing services	Int.	Housing over expenditure net of drugs, CEX 2024	26.8%	26.2%	0.536
β	Discount factor	Int.	Wealth over output, BEA fixed assets	3.00	2.95	0.946
$\xi(j)$	Equivalence scale	Ext.	Square-root scale, OECD	1.41	n.a.	1.41
Block 8: Rental ladder						
h_2	Smallest rental size	Int.	Renters paying over half of income, SCF 2022	19.2%	25.8%	0.0407
h_3/h_2	Middle rung over the smallest	Ext.	Set	4.00	n.a.	4.00
h_4/h_2	Largest rung over the smallest	Ext.	Set	8.00	n.a.	8.00
Block 9: Earnings and employment						
J	Maximum age	Ext.	Age 85	60	n.a.	60
J_K	Years the child lives at home	Ext.	Parent aged 26 to 50	25	n.a.	25
J_R	Retirement age	Ext.	Age 66, SSA full retirement age	41	n.a.	41
$\varepsilon(j)$	Age profile of efficiency	Ext.	Hump peaking at age 56, Greenwood et al. (2025)	Profile	n.a.	Profile
ρ_y	Persistence of productivity	Ext.	Guvenen et al. (2023)	0.90	n.a.	0.90
σ_ζ	Standard deviation of its shocks	Int.	Bottom half's share of earnings, SCF 2022	17.2%	16.2%	0.413
ρ	Persistence of productivity across generations	Ext.	Solon (1992), Mazumder (2005)	0.50	n.a.	0.50
$\bar{\pi}^{eu}$	Job separation rate	Int.	Unemployment rate, BLS CPS 2014–24	4.9%	5.6%	0.0268
$\varpi(y)$	Separations by productivity	Ext.	Unemployment by education, BLS 2019	Profile	n.a.	Profile
$\bar{\pi}^{ue}$	Annual job-finding rate	Ext.	Krueger et al. (2014)	0.55	n.a.	0.55
p_{ui}	Share of separations that receive UI	Ext.	UI applicants among the unemployed, BLS 2022	0.30	n.a.	0.30
Block 10: Baseline survival and SMI onset						
$\bar{\phi}(j)$	Baseline survival by age	Ext.	US life table 2010, Arias (2014)	Profile	n.a.	Profile
$\varsigma_m(j)$	Age profile of SMI onset	Ext.	SMI prevalence by age, NSDUH 2024	Profile	n.a.	Profile
Block 11: Interest and taxes						
r	Risk-free rate	Ext.	Kaplan et al. (2020)	0.03	n.a.	0.03
φ	Tax progressivity	Ext.	Heathcote et al. (2017)	0.85	n.a.	0.85
τ_ℓ	Level of the labour income tax	Int.	Average labour income tax, McDaniel (2007)	22.4%	20.9%	0.894
τ_k	Capital income tax rate	Ext.	US long-run rate on gains and dividends	0.15	n.a.	0.15
Block 12: Social security benefits and programmes' rules						
$S(y)$	Social security	Ext.	SSA primary insurance amount	Statute	n.a.	0.90, 0.32, 0.15
$\underline{S}$	SSI floor over average labour income	Ext.	SSA federal benefit rate	0.177	n.a.	0.177
ψ_{ui}	UI replacement rate	Ext.	Department of Labor, statutory rates	0.50	n.a.	0.50
$\bar{y}$	Cap on the UI earnings base	Ext.	Department of Labor, state benefit maxima	1.00	n.a.	1.00
κ_V	Voucher means test	Ext.	HUD rule	0.50	n.a.	0.50
κ_r	Tenant's rent, share of income	Ext.	HUD rule	0.30	n.a.	0.30
ϱ	Food stamp clawback rate	Ext.	SNAP statute, 7 CFR 273.10	0.30	n.a.	0.30
Block 13: Drug market						
J_E	Maximum age at which use can begin	Ext.	Model age 55, that is age 80	55	n.a.	55
ν_1	Supply shifter	Ext.	Normalisation, so $p_v = 1$	Norm.	n.a.	Norm.

Notes: the Model column is the Housing First benchmark. As in Table 3, I use n.a. whenever a parameter is set externally and thus has no moment to reproduce.

Block 7: Preferences. The four parameters in this block determine what a household wants, and what it is willing to give up to get it.

σ, risk aversion. It sets the marginal utility of consumption, so it affects every welfare comparison in the paper. Increase it and households want smoother consumption and insurance is worth more, the voucher included. I set it to the standard value of two, following Attanasio and Weber (2010).

γ**, the weight on housing services.** It is the weight on housing in the bundle of consumption and housing, so it sets what a dwelling is worth against consumption, and that bundle is what the drug is weighed against. As γ rises housing is worth more relative to consumption, households buy more of it, and the share of spending that goes on housing rises with it. γ is chosen for the share it produces: shelter plus utilities take 26.8 percent of expenditure in the 2024 Consumer Expenditure Survey (CEX), which excludes illicit drug spending.

β**, the discount factor.** It sets how much households discount the future, and thus how much they save and how much the prospect of death deters them from using drugs. As it rises households save more, the wealth-to-output ratio rises and households are more deterred by death. It is chosen so that wealth over output matches the national accounts, a ratio of three, from the Bureau of Economic Analysis (BEA) fixed asset tables against National Income and Product Accounts (NIPA) output, both read at the 2023 vintage, the last year the fixed asset tables cover.

$\xi(j)$**, the equivalence scale.** The household's consumption and dwelling are shared with the child while it lives with its parent, and thus are divided by $\xi(j)$ to give what each member gets. Raise it and each member gets less of a given budget, and the child counts for more of the household, through the weight $\xi(j) - 1$ in (12). It is the square-root scale (OECD, 2013), $\sqrt{N(j)}$, where $N(j)$ in the model is set to two while the child lives with the parent ($j \leq J_K$), from the parent's age 26 to 50, and to one after the child has left ($j > J_K$).

Block 8: Rental ladder. These three parameters build the grid of dwelling sizes a household chooses from. Below the smallest rung is the shelter, h_1, which was calibrated in Block 2.

h_2**, the smallest rental size.** The size of the smallest dwelling is what determines how much of its income a poor household has to spend to avoid homelessness. When the smallest dwelling is larger, it is more expensive to rent, and thus more households are cost burdened and more likely to become homeless, and a voucher is worth more to them. It is chosen so that the fraction of renters paying more than half of their income for housing is similar to the data, 19.2 percent in the 2022 SCF; the model gives 25.8 percent.

h_3/h_2 **and** h_4/h_2**, the spacing of the ladder.** The two ratios here determine the amount of housing services obtained by renting the middle and largest dwelling options. Therefore, they affect what a household that can afford more than the smallest unit can rent. Larger ratios make each step up the ladder cost more to rent, and thus fewer households can afford the larger options. The two larger rungs are set at 4.00 and 8.00 times the smallest.

Block 9: Earnings and employment. This block has eleven parameters that govern the maximum periods of life a household can live, until what age it works, and what it can expect to earn while it is of working age. These parameters also set how persistent and frequent shocks to employment and productivity are.

J**,** J_K **and** J_R**, the length of life, of child rearing and of working life, respectively.** A household lives at most J periods, has its child at home for its first J_K of them and works until model age J_R, after which it retires on social security. J and J_R set the maximum number of years a household can live and work, respectively, and J_K sets how long a parent's drug use reaches its child, since the child lives with them during those J_K years. A higher J or J_R makes a disorder that shortens life or employment more costly, whereas a higher J_K makes the budget and childhood homelessness externalities last longer. I set $J = 60$, so that with $j = 1$ at age 26 the last age is 85, $J_K = 25$, so that the child lives at home while its parent is 26 to 50, and $J_R = 41$, so that retirement begins at age 66, the full retirement age under social security for those born between 1943 and 1954.

$\varepsilon(j)$**, the age profile of efficiency.** This is the deterministic hump in earnings over the life cycle. Without it every household with the same productivity would earn the same at every age, and the ages at which households are poor, and so at risk of homelessness and of using, would be wrong at both ends of working life. I take it as the exponential of a quadratic in age used by Greenwood et al. (2025), $\varepsilon(j) = \exp\{-(j-1)^2/1800 + (j-1)/30\}$ with $j = 1$ at age 26. It peaks at age 56, having risen 65 percent from age 26, and then falls 4 percent from the peak to age 65, the last working age.

ρ_y **and** σ_ζ**, the productivity process.** Log productivity is persistent but not permanent: it follows an autoregressive process with persistence ρ_y and shocks with standard deviation σ_ζ. σ_ζ is what decides how many households are poor enough to be at risk of homelessness, and the share of earnings that goes to the bottom half of working households falls as it rises. I set $\rho_y = 0.90$, the persistence of the income process in Guvenen et al. (2023). σ_ζ is calibrated to the bottom half's share of earnings among households aged 26 to 65 with positive wage income in the 2022 SCF, 17.2 percent.

ρ**, the persistence of productivity across generations.** The child's log productivity at entry into adulthood is drawn around its parent's with weight ρ and around the mean with weight $1 - \rho$, so ρ is how much of its parent's current productivity a child keeps. It is the channel through which anything that lowers a parent's productivity lowers its child's: a parent whose own productivity carries the penalty δ_d, because its own parent had a SUD, also passes a lower draw on to its child, so a parent's SUD lowers productivity beyond the next generation. Raise it and a child's prospects depend more on its parent's decisions, and the *earnings externality* grows. It is set externally at 0.50, between the intergenerational elasticities of earnings in Solon (1992) and Mazumder (2005).

$\bar{\pi}^{eu}$**, the job separation rate.** This parameter determines the chance that an employed household loses its job each year. It is scaled by $\varpi(y)$ in (21), so separations are likelier at low productivity. Raise it and more households lose work each year, so the unemployment rate rises with it. It is chosen to reproduce the aggregate unemployment rate in the Current Population Survey (CPS) of the Bureau of Labor Statistics (BLS) over 2014 to 2024, 4.9 percent, given the job-finding rate set externally below.

$\varpi(y)$**, separations by productivity.** This parameter tilts the separation rate across productivity, so that low-productivity households lose their jobs more often, which is what puts them at a higher risk of homelessness. A steeper gradient exposes low-productivity households to a higher chance of unemployment, and thus a higher risk of homelessness. It is normalised to average one across households, so that $\bar{\pi}^{eu}$ alone sets the aggregate unemployment rate. Each productivity node takes the unemployment odds of the education group it falls in, from the BLS 2019 annual averages of unemployment by educational attainment for ages 25 and over, with the education groups ordered along productivity by their median earnings.

$\bar{\pi}^{ue}$**, the annual job-finding rate.** This parameter is the probability that an unemployed household finds work each year it is unemployed. With the divisors κ_d^{ue} and κ_m^{ue} it sets how long an unemployed household with a SUD or SMI is expected to remain unemployed. Raise it and spells out of work are shorter on average, so fewer households fall behind on the rent. I set it to 0.55 from the annual exit rate Krueger et al. (2014) estimate on matched CPS panels.

p_{ui}**, the share of separations that receive UI.** A household that loses its job draws UI for one year with probability p_{ui} and nothing otherwise. It sets how many of the newly unemployed fall straight to zero earnings, and how costly being unemployed is. I set it to 0.30, since in the BLS supplement to the CPS for 2022 about seven in ten unemployed people who worked in the past year did not apply for UI after losing their job.

Block 10: Baseline survival and SMI onset. These two parameters are the survival and the SMI-onset age profiles.

$\bar{\phi}(j)$**, baseline survival by age.** This is the baseline chance of surviving to the next age, before any multiplier. It is the base that every mortality multiplier in Section 5, for a SUD, homelessness and SMI, scales. Raise it and households expect to live longer, so the life a SUD puts at risk is worth more. I take the profile from the life table for the total population of the United States in 2010 (Arias, 2014), for ages 26 to 85.

$\varsigma_m(j)$**, the age profile of SMI onset.** This determines the shape of onset across ages, normalised to one at age 26, so that ω_m in Table 3 sets its level alone. It is here rather than in Table 3 because, like the age survival profile, it only sets the baseline. What actually affects the impact of Housing First versus Treatment First is ι, the multiplier a SUD puts on SMI onset, which is in Table 3. It is the onset that reproduces a log-linear fit to the share of adults carrying SMI at each age in NSDUH 2024.

Block 11: Interest and taxes.

r**, the risk-free rate.** This parameter sets the return on saving. A higher interest rate improves the return on savings, and thus households accumulate more assets, which help cushion them during spells of unemployment. I set it to three percent, from Kaplan et al. (2020).

φ**, tax progressivity.** This is the exponent on income in the tax schedule of Section 4.3, and it determines the progressivity of the labour income tax. When its value is lower, the tax system becomes more progressive. In that case, households with low incomes keep more of what they earn. It is set to 0.85, in line with Heathcote et al. (2017).

τ_ℓ**, the scale of after-tax labour income.** Given φ, this parameter sets how much of their labour income households keep after tax on average, so raising it lowers the tax liability. It does not move in the experiments: the budget is balanced by π_V instead. It is calibrated to an average labour income tax rate of 22.4 percent, which is what McDaniel (2007) measures for the United States and what Guvenen et al. (2023) use.

τ_k**, the capital income tax rate.** This parameter affects the after-tax return on savings. A higher capital income tax rate reduces the after-tax return to savings, and this discourages households from saving. I set it to fifteen percent, consistent with the statutory United States rate on long-term capital gains and qualified dividends for the middle of the income distribution, which the Internal Revenue Service (IRS) sets out in Topic 409 (`https://www.irs.gov/taxtopics/tc409`).

Block 12: Social security benefits and programmes' rules.

$S(y)$ **and** $\underline{S}$**, social security and its floor.** Retirement income follows the statutory primary insurance amount: it replaces a share of the household's income at its last working age, a share that falls at each of the two bend points of the Social Security Administration (SSA), up to the taxable maximum, with the SSI payment as a floor (the full function is detailed in footnote 16). As $S(y)$ rises the old are richer, so the value of a life still to be lived rises and with it the cost of a disorder that shortens it. The shares $\alpha_1, \alpha_2, \alpha_3$ are $0.90, 0.32, 0.15$ below the first bend point, between the two and above the second, respectively. The bend points k_1 and k_2 and the taxable maximum $y^{\max}$ are the SSA's 2024 dollar amounts divided by its average wage index, 0.22, 1.33 and 2.64, respectively. The floor, $\underline{S}$, is 0.18 of average labour income.

ψ_{ui} **and** $\bar{y}$**, UI.** The first parameter here sets the UI replacement rate, while the second parameter caps the earnings on which UI is paid. Raising either makes the first year out of work less costly for a household who lost their job, and thus makes it less likely that a household would end up in a shelter. I set them to fifty percent and one times average labour income, respectively, taking both from the statutory replacement rates and state benefit maxima that the Department of Labor compiles in its comparison of state UI laws (`https://oui.doleta.gov/unemploy/statelaws.asp`).

κ_V **and** κ_r**, the voucher rules.** These two set the income cutoff for eligibility and the tenant's rent contribution, respectively. Raise κ_V and more households are eligible, so more have a voucher to lose under Treatment First; raise κ_r and a voucher pays less, so losing one costs less. I take both from HUD's own rules: half of median income and thirty percent, respectively. The first is the very low-income limit HUD publishes annually with its income limits (`https://www.huduser.gov/portal/datasets/il.html`); the second is the share of its before-tax income a tenant pays towards the rent, the voucher paying the rest up to the payment standard, under the Housing Choice Voucher programme (`https://www.hud.gov/helping-americans/housing-choice-vouchers`).

ϱ**, the food stamp clawback rate.** This is the rate at which food assistance is withdrawn as income rises. Raise it and food assistance is clawed back faster, so it stops at a lower income and pays less to every household that still receives it. I set it to thirty percent, which is the statutory rule (7 CFR 273.10).

Block 13: Drug market. The two parameters in this block describe the rest of the drug market of Block 6. The first, J_E, fixes who can enter the market (footnote 21); the second is set so that supply equals demand in the benchmark, and is then held fixed.

J_E**, the last age at which use can begin.** This parameter is the last age at which a household without a SUD can start using; one that already has a SUD at that age can keep using. It decides who can

enter the drug market, and so how many households a policy that deters use can reach. As it rises, more old households can start using at little cost, since too little life remains for the disorder to damage (footnote 21), and the model produces too much old-age use. I set it to model age 55, the equivalent of age 80 when the model runs to age 85.

ν_1**, the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark, and kept fixed across experiments, in which the price is free to adjust. It normalises units, not behaviour.